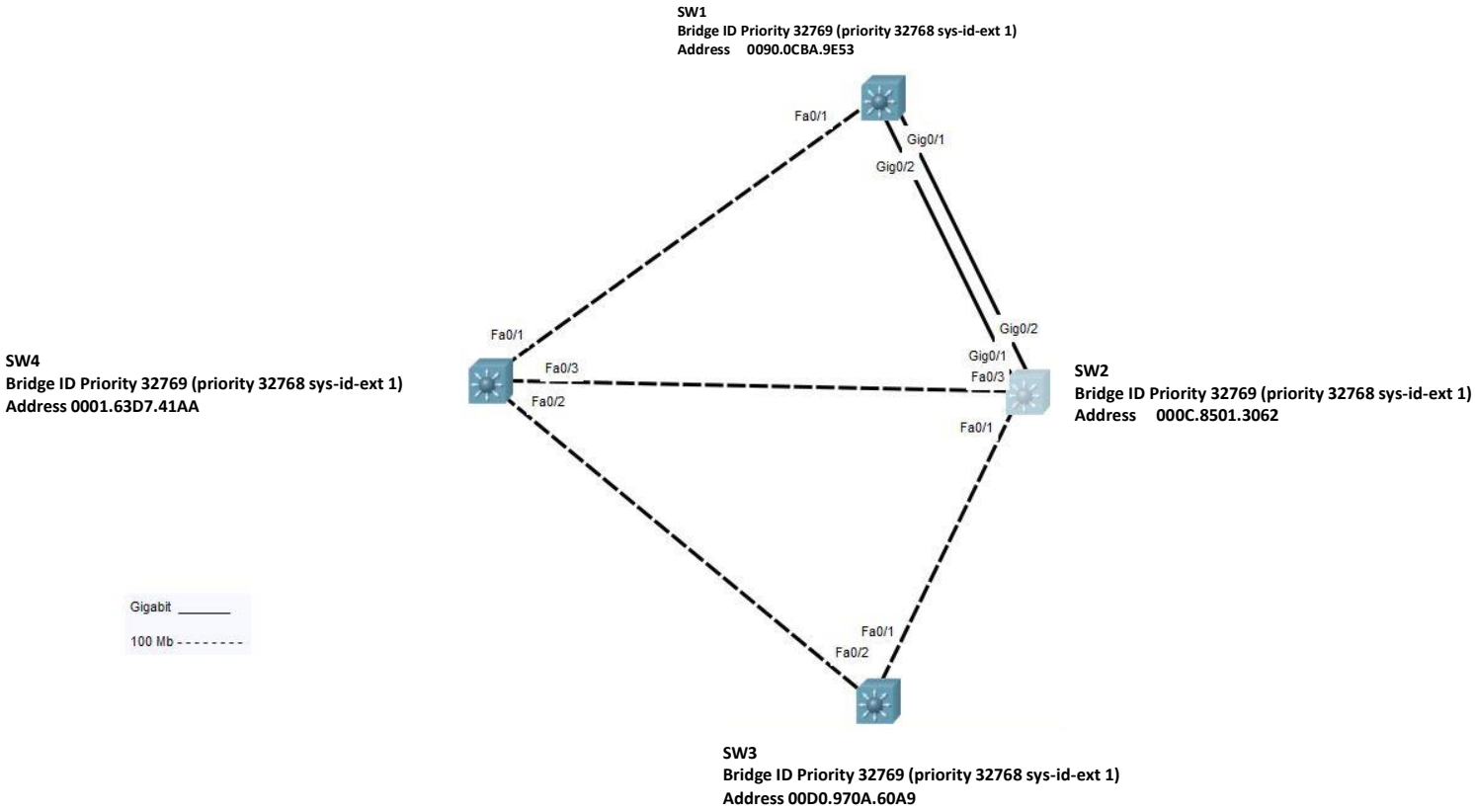
Full Name: _____

Student Number: _____

Part1) STP

Calculate Root Bridge (RB), Root Ports (RP), Designated Ports (DP), and Blocking Ports (BLK) after STP convergence for VLAN 1. (14 points)



	Root Bridge	Root Port	Designated Port(s)	Block Port(s)
SW1				
SW2				
SW3				
SW4				

Part2) Advanced Network Design-PT

Overview

You are the network administrator for UMN. You are responsible for the switched network infrastructure in Owen building. Your team has already connected and configured the server farm correctly. You have 2 departments called ENG and HR each on a separate VLAN.

Your task is to configure the core, distribution and access switches according to the following specifications, so that all the PCs in all subnets get the correct IP Address information from the DHCP Server and can access "**www.inet4011.com**" from the web browser.

- No physical layer changes should be made.
- No configuration changes on the servers in the server farm should be made.
- The servers have been configured with the correct static IP addresses.
- The DNS, DHCP and Web services have already been configured. You can check these configurations on the servers.
- The DHCP server is pre-configured with two pools and allocates default gateways addresses of 10.0.0.1 (for VLAN10) and 13.0.0.2 (for VLAN 13). It also provides the clients with the appropriate DNS setting.
- Routing has been enabled on all switches in campus core and distribution. A dynamic routing protocol has already been configured (EIGRP 100).

Tasks

- Configure the access layer switches (AS1, AS2 and AS3) with the correct VLANs based on the given specification in the network topology. (6 points)
- Enable trunking where appropriate in the Owen building distribution layer. (10 points)
- Set up appropriate L3 Fast Etherchannel between Core1 and distribution switches (DS1, DS2 and DS3) with PAgP channel negotiation protocol. (30 points)
    - For the PT to save correctly, do not use "*no switchport*" on the physical interfaces and check that your configuration is saved correctly when your L3 EtherChannel comes up.
- Configure DS1 to be the primary root bridge for VLAN10, and the secondary root bridge for VLAN13. Configure DS2 to be the primary root bridge for VLAN13 and the secondary root bridge for VLAN10. (10 points)

- Multilayer switches in Owen building distribution (DS1 and DS2) should be configured as the default gateways for VLANs with the following specifications: (10 points)

    - All ENG PCs in VLAN 10 should use DS1 as the default gateway with the default gateways address of 10.0.0.1.

    - All HR PCs in VLAN 13 can use DS2 as the default gateway with the default gateways address of 13.0.0.2.

- Configure the distribution layer devices (DS1, DS2 and DS3) so that the DHCP Server can deliver correct IP Addresses to all PCs. (20 points)

Checking

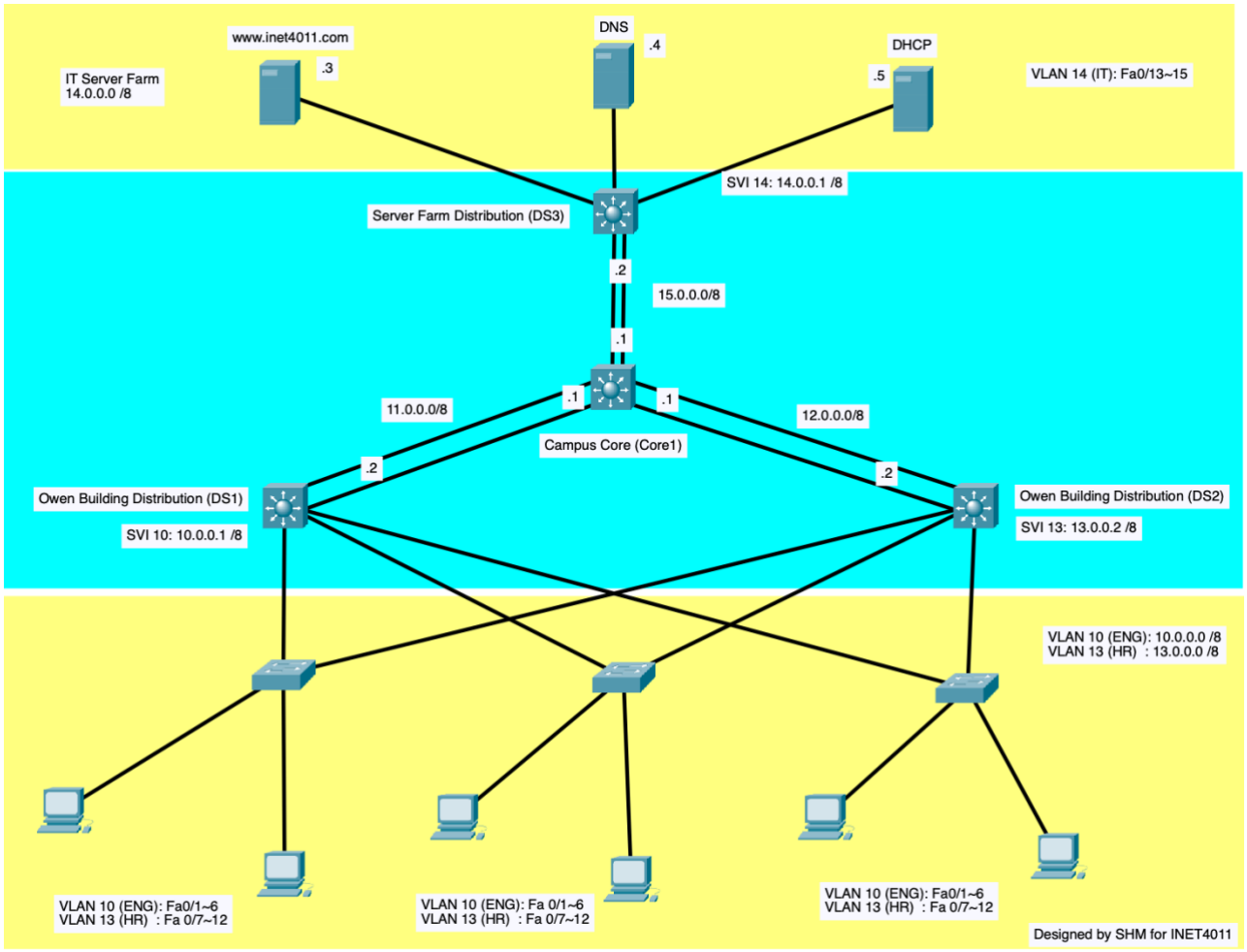
- To check your final configuration make sure you can access "www.inet4011.com" from the web browser app on each PC with a correct DCHP assigned IP address for that department.

- All devices should be able to ping each other.

- All physical links should be UP. Check the actual configuration rather the trusting visual link lights on Packet Tracer.

Submission

- Make sure you save the running configuration of AS1, AS2, AS3, DS1, DS2, DS3 and Core1 by using "write memory" command before saving your PT file and uploading it to Canvas.


www.inet4011.com
.3
DNS
.4
DHCP
.5
IT Server Farm
14.0.0.0 /8
VLAN 14 (IT): Fa0/13~15
SVI 14: 14.0.0.1 /8
Server Farm Distribution (DS3)
.2
15.0.0.0/8
.1
.1
.1
11.0.0.0/8
12.0.0.0/8
Campus Core (Core1)
.2
.2
Owen Building Distribution (DS1)
SVI 10: 10.0.0.1 /8
Owen Building Distribution (DS2)
SVI 13: 13.0.0.2 /8
VLAN 10 (ENG): 10.0.0.0 /8
VLAN 13 (HR) : 13.0.0.0 /8
VLAN 10 (ENG): Fa0/1~6
VLAN 13 (HR) : Fa 0/7~12
VLAN 10 (ENG): Fa 0/1~6
VLAN 13 (HR) : Fa 0/7~12
VLAN 10 (ENG): Fa0/1~6
VLAN 13 (HR) : Fa 0/7~12
Designed by SHM for INET4011

Total Mark: